From FQHE to Fuzzy Sphere Numerical CFT

Landau Levels as a Regulator for 3D Criticality

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[scale=0.7] [ball color=primary!30, opacity=0.6] (0,0) circle (1.5); [primary, thick] (0,0) circle (1.5); ın 0,45,...,315 [accent, -¿, thick] (1.7*cos(,1.7*sin() – (1.2*cos(,1.2*sin(); [primary!70, thick, dashed] (0,0) circle (1.0); [primary] at (0,0.25) $2S$; [primary] at (0,-1.7) monopole flux;

Outline

Haldane Sphere in FQHE

- Particles on S^2 with magnetic monopole flux $2S$
- Lowest Landau level: $\dim \mathcal{H}_{\text{LLL}} = 2S + 1$
- Single-particle basis forms spin-S rep of SO(3)

$$m = -S, \ldots, S \qquad \dim = N_\phi + 1 = 2S + 1 \quad (1)$$

[scale=0.85] [ball color=primary!25, opacity=0.5] (0,0) circle (1.5); [primary, thick] (0,0) circle (1.5); [accent] (0,1.5) circle (0.08) node[right] $+S$; [accent] (0,-1.5) circle (0.08) node[right] $-S$; [primary!50, dashed] (0,0) circle (0.5); [primary!50, dashed] (0,0) circle (1.0); [accent, thick, dashed] (0,0) ellipse (1.5 and 0.4); [accent, font=] at (1.6,-0.5) $m = 0$; [primary] (0,0) circle (0.12); in 0,60,...,300 [primary!60, -¿] (0,0) – (0.6*cos(,0.6*sin();

Key message

The sphere gives a **finite, rotationally symmetric** Hilbert space.

Outline

LLL Projection $\longrightarrow$ Fuzzy Sphere

- After LLL projection, kinetic energy is **quenched**
- Coordinates no longer commute under projection
- Finite matrix algebra approximates functions on S^2

$$[X_i, X_j] \sim \frac{i}{\sqrt{S(S+1)}}\,\epsilon_{ijk}X_k \tag{2}$$

[scale=0.85] [primary, thick] (-3,0) circle (1.2); [primary, font=] **at (-3,-1.7) Classical S^2; [font=] at (-3,-2.2) commuting coords; [-¿, accent, line width=1.5pt] (-1.5,0) – (1.5,0); [accent, font=] at (0,0.5) LLL projection; [ball color=primary!15, opacity=0.4] (3,0) circle (1.2); [primary, thick, dashed] (3,0) circle (1.2); in 2.2,2.6,...,3.8 in -0.8,-0.4,...,0.8 [accent!60] (,) circle (0.04); [primary, font=] at (3,-1.7) Fuzzy S^2; [font=] at (3,-2.2) matrix algebra;**

Key message

Geometry is encoded by **finite matrices**. The sphere becomes “fuzzy”.

Outline

Why This Matters Beyond FQHE

Traditional FQHE use

- Study charge/topological quantum Hall physics
- Interactions *inside* LLL determine phase

Fuzzy sphere CFT use

- Same LLL Hilbert space → **UV regulator**
- Internal flavors encode target field theory

[scale=0.65] [primary, thick, fill=iceBlue!80] (-5,-2) rectangle (-1,2); [primary, font=] **at (-3,2.4) LLL Hilbert Space; [font=, text width=3.5cm, align=center] at (-3,0.5)** $\mathcal{H}_{\mathrm{LLL}}$
$2S+1$ **orbitals;**
[-¿, primary, thick] (-1,1) – (2,1); [font=, primary] at (0.5,1.6) FQHE; [font=, text width=3cm, align=center] at (3.5,1) charge physics
quantum Hall;
[-¿, accent, thick] (-1,-1) – (2,-1); [font=, **accent] at (0.5,-0.4) Fuzzy CFT; [font=**, text width=3cm, align=center] at (3.5,-1) critical spin/flavor sector
charge sector frozen;

[primary!30, thick] (-1,-2) – (-1,2);

Key message

Outline

Constructing a Fuzzy-Sphere Hamiltonian

- Choose LLL creation operators $c^{\dagger}_{m,a}$
- m = orbital index, a = flavor/spin
- Write **rotationally invariant** interactions projected to LLL

$$H = \sum_{\ell} V_{\ell} P_{\ell} + \text{tuning terms} \tag{3}$$

$$H = P_{\text{LLL}} H_{\text{int}} P_{\text{LLL}} \tag{4}$$

[scale=0.75] [draw=primary, fill=iceBlue, rounded corners, text width=7cm, font=, align=center, inner sep=8pt] (box) **Not:** put a lattice CFT Hamiltonian on a sphere

Instead: build a many-body Hamiltonian whose **IR fixed point is the target CFT** ;

Key message

Tune coupling constants to reach a **critical point**.

Outline

1. Starting Point: Haldane Sphere
2. LLL Projection $\rightarrow$ Fuzzy Geometry
3. Beyond FQHE
4. Hamiltonian Construction
5. State-Operator Correspondence
6. Numerical Workflow
7. Why Not a Lattice?
8. Conclusion

State-Operator Correspondence on $S^2 \times \mathbb{R}$

- For $2+1$D CFT on S^2, energy levels $\leftrightarrow$ scaling dimensions
- Angular momentum labels SO(3) spin of operators

$$E_n - E_0 = \frac{\Delta_n}{R} \tag{5}$$

$$\text{states on } S^2 \longleftrightarrow \text{local operators in } \mathbb{R}^3 \tag{6}$$

[scale=0.85] [primary, thick] (0,-2) ellipse (1 and 0.3); [primary, thick] (0,2) ellipse (1 and 0.3); [primary, thick] (-1,-2) – (-1,2); [primary, thick] (1,-2) – (1,2); [accent, -¿, thick] (1.5,0) – (2.5,0); [accent, -¿, thick] (-1.5,0) – (-2.5,0); [font=] at (0,2.5) $S^2 \times \mathbb{R}$; [font=, accent] at (2.8,0.6) $\Delta_n = R(E_n - E_0)$; [font=, primary] at (3.5,1.5) $\mathcal{O}_1$; [font=, primary] at (3.5,0.5) $\mathcal{O}_2$; [font=, primary] at (3.5,-0.5) $\mathcal{O}_3$;

Key message

Exact diagonalization gives an **operator spectrum**, not just a phase diagram.

Outline

Numerical Workflow

[box/.style=draw=primary, fill=iceBlue, rounded corners, font=, align=center, minimum width=2.2cm, minimum height=0.8cm, arrow/.style=-¿, accent, thick] [box] (n1) Choose $N = 2S + 1$; [box, right=0.3cm of n1] (n2) Build LLL basis; [box, right=0.3cm of n2] (n3) Construct H (SO(3)-inv.); [box, right=0.3cm of n3] (n4) Diagonalize by quantum #s;
[arrow] (n1) – (n2); [arrow] (n2) – (n3); [arrow] (n3) – (n4);
[box, below=0.6cm of n2, fill=paleGold] (n5) Tune to criticality; [box, below=0.6cm of n3, fill=paleGold] (n6) Compare with CFT / Bootstrap / QMC;
[arrow] (n4.south) – ++(0,-0.3) -— (n5.north); [arrow] (n5) – (n6);

Key observables

- Energy spectrum

Advanced

- OPE coefficients

Outline

Why Not Just Use a Lattice?

Lattice approach

- Breaks continuous rotation
- Strong microscopic anisotropy
- CFT data muddied by irrelevant operators
- Cubic symmetry $\neq$ full SO(3)

Fuzzy sphere approach

- SO(3) exact at *any* finite size
- CFT data organized by angular momentum
- Finite-size scaling $\sim$ radial quantization
- No lattice artifacts

[scale=0.7] [primary, step=1, thin] (0,0) grid (3,3); [primary, thick] (0,0) rectangle (3,3); [font=] at (1.5,-0.6) Lattice: breaks SO(3);
[-¿, accent, line width=1.5pt] (4,1.5) – (6,1.5); [accent, font=] at (5,2.2) better;
[ball color=primary!20] (8.5,1.5) circle (1.5); [primary, thick] (8.5,1.5) circle (1.5); ın 0,45,...,315 [accent!60] (8.5+1.2*cos(,1.5+1.2*sin() circle (0.05); [font=] at (8.5,-0.6) Fuzzy S^2: SO(3) exact;

Key message

Outline

Take-Home Message

[box/.style=draw=primary, fill=iceBlue, rounded corners, font=, align=center, inner sep=10pt, minimum width=3.5cm] [box] at (-3.5,0) (b1) FQHE
supplies the Hilbert space
LLL on Haldane sphere; [box] at (0,0) (b2) Fuzzy Sphere
supplies the regulator
Finite noncommutative S^2; [box] at (3.5,0) (b3) Numerical CFT
supplies the goal
Conformal spectra & data;
[-¿, accent, thick] (b1) – (b2); [-¿, accent, thick] (b2) – (b3);

[draw=accent, fill=paleGold, rounded corners, font=, align=center, inner sep=12pt] **The surprise:**
A tool born in quantum Hall physics
becomes a **microscope for 3D conformal field theory** ;

Thank You

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